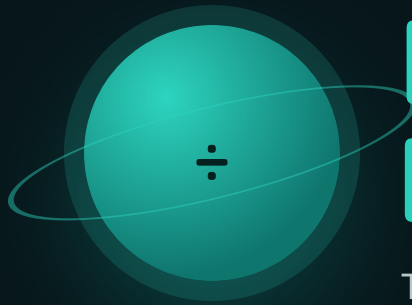




Long Division

Two-digit divisors, and a remainder that means something.

THE BIG QUESTION

When you divide and there's something left over, when do you round up, round down, or keep the remainder?

HANDS-ON PROJECT

Road Trip Budget

Plan a real route: miles per day, gallons per tank, dollars per person. Every number divided by something.

FLUENCY GAME

Remainder Race

Roll a divisor, name the remainder before your opponent does. Wrong answers cost a turn.



AXIOM • MISSION COMMANDER, NINE TOURS

“Three buses for 100 people means three, not 3.33. The arithmetic gives you a number — the situation tells you what to do with it.”

Pre-Assessment

Twelve items, six strands. Anything you already own, we skip.

This is not a test. Nothing here counts for a grade. Item F1 has no single right answer — it wants a decision and a reason.

STRAND A
DIVISION FACTS

A1 $84 \div 12$

.....

A2 $72 \div 8$

.....

STRAND B
ONE-DIGIT
DIVISOR

B1 $372 \div 3$

.....

B2 $618 \div 6$

.....

STRAND C
TWO-DIGIT
DIVISOR

C1 $936 \div 24$

.....

C2 $4,536 \div 21$

.....

STRAND D
ZEROS AND
ESTIMATES

D1 $7308 \div 36$

.....

D2 Estimate $3120 \div 39$

.....

STRAND E
REMAINDERS

E1 $100 \div 7$, with the remainder

.....

E2 $50 \div 8$, with the remainder

.....

STRAND F
INTERPRETING

F1 100 people, buses hold 30. How many buses?

.....

F2 \$50 split 4 ways, to the cent

.....

Skip Chart

Score the pre-assessment, then cross days off the four-week calendar.

2 / 2 → SKIP

1 / 2 → COMPRESS

0 / 2 → TEACH

STRAND	ITEMS	COVERS	SCORE	IF 2/2, REMOVE
A • Division facts	A1 A2	Every Warm-Up tier	--- / 2	Warm-Up drops to 3 items all mission.
B • One-digit divisor	B1 B2	Week 1 • lesson 1.1	--- / 2	Day 1.1. B2 has an interior zero, so a pass there matters more than B1.
C • Two-digit divisor	C1 C2	Week 1 • lessons 1.2 to 1.4	--- / 2	Compress Week 1 to two days, but keep 1.3 — estimating the quotient is the skill everything else rests on.
D • Zeros and estimates	D1 D2	Week 2 • lessons 2.2 and 2.3	--- / 2	Day 2.2 only. A zero in the quotient is the single most common silent error in long division.
E • Remainders	E1 E2	Week 3 • lesson 3.1	--- / 2	Day 3.1. Finding a remainder is easy; deciding what it means is not.
F • Interpreting	F1 F2	Week 3 • lessons 3.2 to 3.4	--- / 2	Nothing. This is the Big Question — teach it even at 2/2.

ONE NOTE BEFORE YOU START

Watch F1 closely. A student who answers 3.33 has done the arithmetic correctly and missed the question — 100 people need four buses. That gap between a correct calculation and a correct answer is what Week 3 exists to close, and it is worth more than any amount of extra long-division drill.

One Digit, Cleanly

Divide, multiply, subtract, bring down. Say the four steps out loud.

Warm-Up FACTS YOU ALREADY OWN

$84 \div 12$

$72 \div 8$

$63 \div 9$

$120 \div 4$

$300 \div 5$

$360 \div 6$

372 ÷ 3 · THE FOUR STEPS, TWICE

$$\begin{array}{r} 124 \\ 3 \overline{) 372} \\ \underline{- 3} \\ 07 \\ \underline{- 6} \\ 12 \\ \underline{- 12} \\ 0 \end{array}$$

1 • How many 3s in 3? One. Write it above the 3.

2 • $1 \times 3 = 3$. Write it below.

3 • $3 - 3 = 0$.

4 • Bring down the 7.

Then repeat: how many 3s in 7? Two, with 1 left. Bring down the 2 to make 12, and $12 \div 3 = 4$ exactly.

Estimate first: 372 is close to 360, and $360 \div 3 = 120$. The answer should be just over 120.

Core ESTIMATE ABOVE EACH ONE FIRST

1. $372 \div 3$

.....

2. $618 \div 6$

.....

3. $896 \div 8$

.....

4. $1,236 \div 4$

.....

5. In item 2 the quotient has a zero in it. Which of the four steps produces that zero?

.....

Challenge NOT GRADED

C1. $6,036 \div 6$. There are two zeros in the answer. Where do they come from?

.....

C2. Multiply your answer to item 4 by 4. You should get 1,236 back. Why does that check work?

.....



AXIOM: Multiply your answer back. Division is the only operation that hands you a free check, and it takes ten seconds.

Two-Digit Divisors

Round the divisor to guess, then check the guess and adjust.

Warm-Up MULTIPLES OF THE DIVISOR

24×10

24×20

24×30

24×40

24×5

24×9

Write these six down before you start today's Core. A list of the divisor's multiples turns guessing into looking up.

936 ÷ 24 • GUESS, CHECK, ADJUST

$$\begin{array}{r} 39 \\ 24 \overline{) 936} \\ \underline{- 72} \\ 216 \\ \underline{- 216} \\ 0 \end{array}$$

Round 24 to 20. How many 20s in 93? About four — but $24 \times 4 = 96$, too big. Try three: $24 \times 3 = 72$. That fits.

Bring down the 6 to make 216. How many 24s? Round again: 20 into 216 is about ten, but $24 \times 10 = 240$, too big. Try nine: $24 \times 9 = 216$ exactly.

The first guess being wrong is normal. Rounding the divisor down makes your guess too big; rounding it up makes it too small. Expect to adjust.

Core

1. $936 \div 24$

.....

2. $672 \div 21$

.....

3. $884 \div 34$

.....

4. $1,104 \div 46$

.....

5. For item 3 you rounded 34 to 30. Did that make your first guess too big or too small? Why?

.....

Challenge

C1. $812 \div 29$. Rounding 29 to 30 makes this quick — but which way will your guess be wrong?

.....

C2. Which is easier to divide by, 25 or 29? Both are close to 30, so say what actually makes the difference.

.....

Estimate the Quotient

Know roughly how big the answer is before you start dividing.

✦ Warm-Up ROUND BOTH, THEN DIVIDE

$$400 \div 20$$

$$600 \div 30$$

$$900 \div 30$$

$$1200 \div 40$$

$$2000 \div 50$$

$$3600 \div 60$$

HOW MANY DIGITS WILL THE ANSWER HAVE?

$$4536 \div 21$$

$21 \times 100 = 2,100$ — fits.

$21 \times 1000 = 21,000$ — too big.

So the answer is in the hundreds.

$$812 \div 29$$

$29 \times 10 = 290$ — fits.

$29 \times 100 = 2,900$ — too big.

So the answer is in the tens.

Why it matters

If you expect a three-digit answer and get two, you dropped a digit somewhere. The estimate catches it before you finish.

◆ Core ESTIMATE FIRST, THEN DIVIDE, THEN COMPARE

1. $4,536 \div 21$

2. $3,120 \div 39$

3. $8,464 \div 46$

4. $2,451 \div 43$

5. For each item above, write how many digits you expected in the answer. Did any of them surprise you?

★ Challenge

C1. Somebody divided 4,536 by 21 and got 26. How does the digit count tell you they went wrong without checking any arithmetic?

C2. $12,060 \div 60$. Estimate the number of digits, then find the answer. The zero in the middle is not an accident.

Check by Multiplying

Quotient times divisor, plus the remainder, equals what you started with.

✦ Warm-Up MULTIPLY BACK

$$39 \times 24$$

$$32 \times 21$$

$$26 \times 34$$

$$24 \times 46$$

$$14 \times 7 + 2$$

$$6 \times 8 + 2$$

THE CHECK, WRITTEN OUT

$$100 \div 7 = 14 \text{ r } 2$$

$$14 \times 7 = 98$$

$$98 + 2 = 100 \checkmark$$

Quotient $\times$ divisor + remainder = dividend. If that does not come out, one of the three numbers is wrong.

The remainder must always be **smaller than the divisor**. If you get $100 \div 7 = 13 \text{ r } 9$, the remainder is bigger than 7, which means 13 was too small a quotient. That rule alone catches most remainder mistakes.

◆ Core DIVIDE, THEN WRITE THE CHECK UNDERNEATH

1. $100 \div 7$

2. $500 \div 23$

3. $1,000 \div 31$

4. $750 \div 24$

5. Somebody says $500 \div 23 = 21 \text{ r } 25$. Without dividing, how do you know that is wrong?

★ Challenge

C1. A division has divisor 24 and remainder 7. What is the largest the remainder could have been?

C2. Find a number that leaves remainder 3 when divided by 8, and remainder 1 when divided by 5.

Remainder Race

Name the remainder before your opponent does.

How to play: one player names a number under 200. The other rolls a die for the divisor. Both race to say the remainder out loud — quotient not needed. First correct answer takes the round; a wrong answer costs you the next turn. If the remainder is zero, say “clean” instead.

✦ Warm-Up • remainders only

$17 \div 5$

$23 \div 4$

$50 \div 6$

$100 \div 3$

◆ Play eight rounds

ROUND	NUMBER	DIVISOR	REMAINDER	WON?
1				
2				
3				
4				
5				
6				
7				
8				

★ Challenge

C1. Dividing by 6, how many different remainders are possible? List them.

.....

C2. Which divisor gave you the fastest answers, and why? It is not the smallest one.

.....

Weeks 2–4

Every day below has five graded sets waiting in Practice Bay.

QUIZ • end of week 2

TEST • end of week 4

Week 2 • Bigger Dividends QUIZ FRIDAY

Four-digit dividends, and the zero in the quotient that everybody drops.

MON 2.1

Four digits by two

Where the first digit of the quotient goes, and why.

TUE 2.2

Zeros in the quotient

The place that gets skipped is the place people lose.

WED 2.3

Estimate the quotient

A rough answer means a wrong one gets noticed.

THU 2.4

Divisors near a ten

Rounding 29 to 30 makes the guessing quick.

FRI

Mid-unit quiz

Page 10, then the Road Trip Budget begins.

Week 3 • What the Remainder Means

The mission's real content. Same division, four different right answers depending on what you asked.

MON 3.1

Find the remainder

It is always smaller than the divisor. Always.

TUE 3.2

Round up or down

Buses round up. Full boxes round down.

WED 3.3

Keep it as a fraction

Pizza does not round. Neither does money.

THU 3.4

Four questions, one division

$100 \div 7$ answered four ways, all correct.

FRI

Budget the miles

Miles per day, tanks per trip, dollars per person.

Week 4 • Proof UNIT TEST

The Road Trip Budget gets finished, and every rounding decision in it gets defended.

MON 4.1

Finish the budget

Every line item divided out and totalled.

TUE 4.2

Defend the rounding

Pick three lines and say why each rounded the way it did.

WED 4.3

Mixed review

Long division and remainder interpretation together.

THU

Error journal sweep

Fix only what repeats.

FRI

Mission 03 test

Pages 11–12. Twelve items plus one explanation.

Mid-Unit Quiz

Eight items from Weeks 1 and 2. 85% to keep flying — that's 7 out of 8.

1. $372 \div 3$

.....

2. $936 \div 24$

.....

3. $4,536 \div 21$

.....

4. $7,308 \div 36$

.....

5. Estimate $3,120 \div 39$

.....

6. $812 \div 29$

.....

7. $100 \div 7$, with the remainder

.....

8. Somebody says $500 \div 23 = 21 \text{ r } 25$. Explain how you know that is wrong without doing the division.

.....

.....

Marking note: item 8 wants the rule, not the right answer. Accept anything saying the remainder cannot be larger than the divisor.

Unit Test • Part 1

Items 1–8. No time limit. Check each answer by multiplying back.

1. $618 \div 6$

2. $884 \div 34$

3. $8,464 \div 46$

4. $9,072 \div 24$

5. $6,036 \div 6$

6. $1,178 \div 31$

7. $750 \div 24$, with the remainder

8. Estimate $6,300 \div 68$ to the nearest ten

Unit Test • Part 2

Items 9–12 and the Big Question.

9. 100 people are going on a trip. Each bus holds 30. How many buses are needed?

.....

10. 100 pencils go into boxes of 30. How many *full* boxes are there?

.....

11. \$100 is shared equally between 30 people. How much does each get, to the cent?

.....

12. A 1,200-mile trip is driven over 5 days. If each day is the same distance, how far per day? If the driver wants no day over 200 miles, how many days does the trip take?

.....

EXPLAIN YOUR THINKING • THE BIG QUESTION

When you divide and there's something left over, when do you round up, round down, or keep the remainder?

Items 9, 10 and 11 are the same division. Use them as your example. Scored on reasoning, not neatness.

.....
.....
.....
.....

11–12

Gold band. The Quotient Badge is earned.

9–10

Silver band. One reteach day, then the badge.

≤ 8

Bronze. Reteach the weak strand and retest that part only.

Key A

Pre-assessment, Lesson 1.1 and Lesson 1.2.

Mark the reasoning, not the handwriting. Any correct method counts.

Pre-assessment

A1 • 7 | A2 • 9 | B1 • 124 | B2 • 103

C1 • 39 | C2 • 216 | D1 • 203 | D2 • 80

E1 • 14 r 2 | E2 • 6 r 2 | F1 • 4 buses | F2 • \$12.50

F1 is the one that matters. An answer of 3 or 3.33 means the arithmetic worked and the question did not get read — mark it wrong and teach Week 3 in full.

Lesson 1.1 • One Digit, Cleanly

Warm-Up • 7 | 9 | 7 | 30 | 60 | 60

Core • 1 • 124 | 2 • 103 | 3 • 112 | 4 • 309

Item 5 • the divide step — 6 goes into 1 zero times, so a zero is written above before bringing the next digit down. C1 • 1,006; the zeros come from 6 not going into 0 and not going into 3. C2 • because multiplication undoes division, so the check recovers the dividend exactly when there is no remainder.

Lesson 1.2 • Two-Digit Divisors

Warm-Up • 240 | 480 | 720 | 960 | 120 | 216

Core • 1 • 39 | 2 • 32 | 3 • 26 | 4 • 24

Item 5 • too big, because dividing by a smaller number gives a larger quotient. C1 • 28; rounding 29 up to 30 makes the first guess too small. C2 • 25, because its multiples are familiar (25, 50, 75, 100) — closeness to a ten is not the only thing that matters.

Key B

Lesson 1.3, Lesson 1.4 and Friday's game.

Where a question asks for a sentence, no sentence means no mark — even with the number right.

Lesson 1.3 • Estimate the Quotient

Warm-Up • 20 | 20 | 30 | 30 | 40 | 60

Core • 1 • 216 | 2 • 80 | 3 • 184 | 4 • 57

Item 5 • three, two, three, two digits respectively. C1 • a two-digit answer where three were expected means a digit was dropped — 21×26 is only about 550, nowhere near 4,536. C2 • 201, and the zero appears because 60 does not go into 6 once the hundreds are taken.

Lesson 1.4 • Check by Multiplying

Warm-Up • 936 | 672 | 884 | 1104 | 100 | 50

Core • 1 • 14 r 2 | 2 • 21 r 17 | 3 • 32 r 8 | 4 • 31 r 6

Item 5 • the remainder 25 is larger than the divisor 23, so another whole 23 could still be taken out. C1 • 23, since the remainder must stay below the divisor. C2 • 11 works, and so does 51 — any number of the form $40n + 11$.

Friday • Remainder Race

Warm-Up • 2 | 3 | 2 | 1

C1 • six remainders — 0, 1, 2, 3, 4, 5. Dividing by n always gives n possible remainders. C2 • usually 5 or 10, because the last digit tells you the answer immediately. Worth drawing out: 2 is fast for the same reason.

Key C

Mid-unit quiz, unit test, and the error journal sheet.

Quiz is out of 8, test out of 12. 85% is 7/8 and 11/12.

Mid-unit quiz

- 1 • 124
- 2 • 39
- 3 • 216
- 4 • 203
- 5 • 80
- 6 • 28
- 7 • 14 r 2
- 8 • remainder rule

Unit test

- 1 • 103
- 2 • 26
- 3 • 184
- 4 • 378
- 5 • 1,006
- 6 • 38
- 7 • 31 r 6
- 8 • about 90
- 9 • 4 buses
- 10 • 3 full boxes

Items 11, 12 and the Big Question

11 • \$3.33 each, with 10 cents left over — accept \$3.33 with a note about the remainder. **12** • 240 miles a day over 5 days; capping at 200 miles means 6 days, since $1,200 \div 200 = 6$ exactly. **Big Question** • full marks for an answer that ties the decision to the situation rather than the number: you round up when everything must be carried (buses), round down when only complete groups count (full boxes), and keep the remainder when the thing can be split (money, distance). Items 9, 10 and 11 are all $100 \div 30$, which is the whole point — if a student notices that unprompted, say so.

Error journal • Mission 03

ONE LINE PER MISTAKE • STUDENT PAGE

Write what went wrong, not what the right answer was. On Thursday of Week 4, read the whole list and fix only what appears twice.

DAY WHAT I GOT WRONG, AND WHY
